
NEXOSxLPIN

Open Ecosystem Development Plan

Branching the verification workbench beyond roadside surveillance
into dynamic civic topics — with full transparency, skill citations,
3D refinement path, and an open iteration model anyone can run.

Version 1.0 · 26 July 2026 · Designed for open distribution
America First · Truth-Seeking · Evidence over narrative
Prefer primary records over posts. Prefer instruments over headlines.
Never launder uncertainty into certainty.

1. Purpose of this document

This plan explains how the full NEXOSxLPIN + LPIN skill stack can move beyond its first domain (roadside surveillance / ALPR) into any dynamic civic topic while preserving evidence discipline. It is written for open distribution: every claim about architecture is traceable to the public skill and product surfaces, and every contribution path is explicit.

The stack has two permanent layers: (1) the skill / governance layer (LPIN / AOS v3.0) that enforces Evidence / Inference / Assumption separation, Layer-0 gates, and multi-agent orchestration; and (2) the runnable verification workbench that turns those rules into desks, claim ledgers, maps, and illustrative 3D scenes. The workbench is deliberately domain-swappable.

2. Non-negotiable discipline (must survive every branch)

Any fork or topic adaptation is only valid if these remain intact:

- **Evidence-gated claims** — Funding, jurisdiction, privacy, and density statements appear only after primary or strong secondary sources (or live public APIs) are bound. No invented numbers or award IDs.
- **Tri-state + honesty flag** — +1 Supported, 0 Not proven, -1 Disputed, plus an explicit “plausible / unverified” treatment. No soft intermediate scores.
- **Layer-0 export gates** — High-stakes actions (export, ladder promotion, pack replace) stay blocked while unresolved -1 claims exist.
- **Progressive disclosure** — State before City; Design Lab as a strict cascade; advanced Forge/Massing controls unlock after verification loops.
- **Explicit export only** — Nothing auto-downloads. Oversight / publish packs are deliberate operator actions.
- **Illustrative geometry only** — 3D objects are stand-ins derived from scored claims. They are never presented as forensic reconstructions.

3. Relevant GitHub skills and how they map

The permanent governance and capability layer lives in the LPIN skill repository. Primary public reference: <https://github.com/AdventureNLearn/AOS-v3---LPIN> (portable skills/ structure). Below are the skills most directly exercised when the workbench branches into new civic topics.

Skill	Role in branching topics	Typical activation
evidence-gate	Forces +1 / 0 / -1 labeling; blocks export on open -1	Every claim score and export path
sovereign-lens	Module-0 routing + narrative integrity; optional psyop-narrative-detector layer	High-stakes or media-adjacent desks
4-agent-orchestration	Structured multi-agent hand-offs with role separation	Complex multi-jurisdiction or multi-source audits
mission-spine-guard	Real-time alignment to permanent identity tokens and truth-seeking primacy	All Tier-1 outputs
civic-intelligence-coordinator	Coordinates permit, public-records, oversight, and jurisdiction work	Any civic domain fork
public-records-forensics	Document-level analysis of permits, contracts, minutes, resolutions	Source binding and ledger hygiene
jurisdiction-ops	Knowledge graphs, schema validation, regulatory corpora	State/city gating and density services
permit-coordinator	Regulatory pathway engine; inspection frameworks	Installation-condition matrices
oversight-kit-builder	Packages evidence bundles and audit kits for citizen accountability	Export Kit / Oversight Kit generation
influence-mapping-analyst	Entity and relationship mapping with evidence trails	Actor-claim-source graphs
gis-layer	Leaflet/map layers, public spatial data, pin management	Map pins and spatial desks
visual-systems-architect	Mermaid, charts, status-aware visuals, evidence-legible diagrams	Status boards, timelines, scene legends
glyphos	Layered symbolic encodings and visual operators	Status rail / badge language
working-doc-manager	Durable decisions, verification log, latent state across sessions	All multi-step topic forks
ops-hardening-architect	Audits, verification gates, fail-fast quality	lint / test / build / smoke gates

Citation note: Skill names and roles above correspond to the portable SKILL.md files under the AOS-v3---LPIN repository and the local custom skill tree. Descriptions are functional summaries; the authoritative text is always the SKILL.md itself.

4. Branching beyond roadside surveillance

The first shipping domain is roadside surveillance devices (ALPR and related). The same workbench is intentionally a template. Domain depth lives in data packs, catalogs, engines, and labels — not in new product brands or forks of the governance layer.

4.1 What stays fixed

Stack (Vite / React / TypeScript / Zustand / map + 3D surface), tile/workspace modes, claim ledger + multi-loop verify, Layer-0 export path, status visual language, and the five non-negotiables in Section 2.

4.2 What is swapped per topic

- Product / engineering catalog (devices, models, mounting conditions).
- Installation-condition matrix and public-asset conflict lists.
- Audit and privacy engine rules (findings, ladder, retention, grant keywords).
- Density / spatial service tags (Overpass or topic-appropriate public source).
- Research card templates, Log prefixes, sample jurisdiction.
- Optional SME lenses and training desks.

4.3 Ready topic families (illustrative)

Police body-worn cameras · Automated traffic enforcement · Municipal CCTV · Public facial recognition · Municipal UAS / drone programs · Smart streetlight multi-sensor poles · Public Wi-Fi / kiosk data collection · Environmental / air-quality sensor networks · Transit AVL / passenger counting · School-zone safety monitoring. Each family reuses the identical install path, workspace, and trust model.

New topics are added by following the same sequence: rename surface labels → replace catalogs → rewrite domain engines → update spatial tags → keep governance and gates untouched → re-run lint / test / build / smoke. Fidelity is measured by the checklist in Section 7.

5. 3D modeling: current state and required polish

Procedural Forge and Massing Viewer already turn claim language into readable 3D stand-ins (vehicle, path, crowd, docket stack, vessel, debris, building massing, etc.). Object selection is driven by nouns in scored claims, ranked by importance, and tagged with verifiability flags. Accent colors follow claim status. This is intentionally illustrative — never forensic.

5.1 Gaps that still reduce investigative insight

- **Claim-to-object fidelity** — Noun matching can still surface generic or near-duplicate objects. Ranking and near-duplicate removal need tighter thresholds so only story-critical objects remain.
- **Status legibility at distance** — In Full-scene mode, status color on small objects can be hard to read. Materials, emissive rims, and label density need refinement so +1 / 0 / -1 / plausible remain unmistakable.
- **Reasoning panel depth** — Solo view should surface the exact claim text, score, source-binding state, and short “why this object” rationale without requiring a separate Claims pane.
- **Temporal and relational scenes** — Current meshes are mostly static. Investigative value rises when path strips, sequence markers, or contested-locus pairs can express before/after or competing claims.
- **Honesty labeling** — “Illustrative — derived from scored claims / not forensic” must remain persistent in both Full and Solo modes, including exported stills.
- **Performance and clarity under load** — Desks with many objects need automatic level-of-detail and collision-free layout so the scene stays readable rather than decorative.

5.2 Refinement targets (map to investigative points)

Every 3D improvement must increase an operator’s ability to answer: What is supported? What is only plausible? What is disputed? What is still missing? Concrete targets:

- Status-linked materials and rims that survive small-object Full-scene views (align to UI Supercharge Spec tokens).
- Solo-object reasoning panel that always shows linked claim, score, and source state.
- Contested-locus pair view (same object under 0 vs -1 or +1 vs plausible) for rapid visual comparison.
- Stricter objectReasoning ranking so only high-importance, non-duplicate nouns seed the scene.
- Optional light temporal markers (sequence indices) when claims carry ordered events.
- Persistent illustrative disclaimer on every scene export.

Implementation anchors already named in product briefs include `src/lib/forge/objectReasoning.ts`, `sceneMeshes.ts`, `generators.ts`, and the `Massing / ProceduralForge` modules. Visual language is governed by the UI Supercharge Spec (Claim Status system).

6. Open iteration and development plan

The ecosystem is intended to be refined by anyone who accepts the non-negotiables. The following plan is the transparent operating model.

6.1 Principles for contributors

- Evidence over narrative — every new claim path must remain scoreable and export-gated.
- Portable over branded — domain depth lives in packs and catalogs, not new product names.
- Explicit over ambient — no auto-download, no hidden state, no soft scores.
- Replicable over tribal — SKILL.md files, working documents, and quality gates must be enough for a stranger to reproduce results.
- Human locus of responsibility — Layer-0 and working-document trails keep accountability with the operator, not the model.

6.2 Contribution surfaces

Skills (LPIN repo) — New or improved SKILL.md files under the portable skills tree; must declare purpose, triggers, integration points, and OPSEC notes. Citation: <https://github.com/AdventureNLearn/AOS-v3---LPIN>

Workbench domain packs — Catalogs, installation matrices, audit/privacy engine rules, spatial tags, and sample desks for a new civic topic. Must pass the fidelity checklist.

Visual / 3D refinements — Status materials, objectReasoning ranking, Solo reasoning panels, contested-locus views, and performance improvements that preserve the illustrative disclaimer.

Verification and hardening — Additional smoke tests, lint rules, and acceptance checks that make Layer-0 and score hygiene harder to bypass accidentally.

6.3 Recommended iteration cadence

- **Gate 0 — Intent lock** — State the topic, the non-negotiables that will be preserved, and the success metric (usually “operator can score claims and export only when clean”).
- **Gate 1 — Catalog + engine** — Swap products, conditions, audit/privacy rules, and spatial tags. No governance changes.
- **Gate 2 — Visual parity** — Confirm claim status language, map pins, and scene objects still obey the locked color and honesty rules.
- **Gate 3 — Quality gates** — npm test, lint, build, and domain smoke must exit 0. Manual acceptance checklist must pass.
- **Gate 4 — Working document** — Log decisions, evidence items, and any remaining open –1 issues. Export only after Layer-0 clear.
- **Gate 5 — Package** — Zip with the same install/start pattern so a new operator can reproduce the path in one session.

6.4 Quality gates anyone can run

From the product root (example path C:\NEXOSxLPIN):

```
npm.cmd run test
npm.cmd run lint
npm.cmd run build
node scripts\smoke-sme-congress.mjs (or topic-equivalent smoke)
```

Additional manual checks: State→City unlock works; Design Lab cascade does not freeze; public-asset conflicts remain visible; Export Kit is button-only; every claim remains Evidence / Inference / Assumption; 3D scenes carry the illustrative disclaimer.

7. Fidelity checklist for any domain fork

1. Build exits 0; lint and tests clean on the active tree.
2. Sample jurisdiction produces layout + live Log lines from public or bound sources.
3. Design Lab cascade never freezes and never creates sticky massing loops.
4. State → City (or equivalent) unlocks density / spatial service.
5. Public-asset or topic-equivalent conflict lists remain visible and human-readable.
6. Export / Oversight Kit is deliberate and Layer-0 aware.
7. Claim status visual language (+1 / 0 / –1 / plausible) is consistent across ledger, map, and scene.
8. 3D objects remain illustrative; disclaimer present; no forensic presentation.
9. Working document or equivalent audit trail records decisions and open issues.

10. No secrets or personal data in sample packs; safe-URL checks on external links.

8. Transparency and open distribution posture

This document and the referenced skill repository are intended to be shared. The goal is replicable high-fidelity verification, not proprietary lock-in. Operators and contributors are expected to:

- Keep Layer-0 and tri-state discipline intact in every public fork.
- Cite the skill names and product surfaces they rely on.
- Prefer primary public records and instruments over secondary narrative sources.
- Label illustrative geometry clearly so no reader confuses it with surveyed fact.
- Publish working-document excerpts or verification logs when claiming results.

Training desks and sample packs are not legal advice; scores are operator hygiene, not judicial findings. Export actions remain the operator's responsibility.

The stack is ready to branch. The discipline is not optional. Anyone can refine catalogs, engines, visuals, and tests; no one is free to soften the evidence gates.

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Primary skill repository: <https://github.com/AdventureNLearn/AOS-v3---LPIN>